EEE446 ELECTRONIC MAITENANCE AND REPAIRS NOTES I

COMPILED BY

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Syllabus

Electronics Maintenance and Repairs

PROGRAMME: HND IN ELECTRICAL/ELECTRONIC ENGINEERING TECHNOLOGY (ELECTRONICS AND TELECOMMUNICATIONS OPTION)			
Course: ELECTRONIC MAINTENANCE AND REPAIRS		Course Code: EEE 446	Contact Hours: 0/0/4
Course Specification: Theoretical Content & Practical Content			
	General Objective 1.0: Know the use of various electrical and electronic measuring instruments		
WEEK	Specific Learning Outcome	Teachers Activities	Resources
1 - 6	1.1 Use the oscilloscope to measure voltage, current and wave-forms of an electronic circuit. 1.2 Use the oscilloscope in tracing faults in electronic circuits. 1.3 Use the signal generator to inject test signals into a given circuit, e.g. into amplifier circuit. 1.4 Use multimeters (analogue and digital) to measure such quantities as: a. voltage; b. current; c. resistance; d. out-put power; e. transistor parameters. 1.5 Use the capacitance tester to measure the value of a capacitor. 1.6 Use the transistor tester to: a. identify the terminal of a transistor; b. identify the type of transistor; c. measure the parameter/characteristics of a transistor; d. measure the forward and reverse resistance of a diode. 1.7 State the safety precautions to be taken during testing of equipment.	- Ask students to perform measurement	

PROGRAMME: HND ELECTRICAL/ELECTRONICS TECHNOLOGY			
Course: DIGITAL COMMUNICATION III		Course Code: EEE 445	Contact Hours: 2/0/3
Course Specification: Theoretical Content			
	General Objective 2.0: Know how to diagnose faults and rectify them in electronic equipment		
WEEK	Specific Learning Outcome	Teachers Activities	Resources
7 - 12	2.1 Explain the testing methods necessary to locate the following faults in electronic circuits: a. Short circuit; b. Open circuit. 2.2 Explain the testing methods necessary to locate the following faults in electronic circuits: a. Open circuit; b. Short circuit. 2.3 Diagnose open circuit, short circuit and other faults in electronic circuits and equipment using appropriate instruments. Examples of equipment are TV (black and while & colour) video sets, radio sets, tape recorders, telecommunications equipment, audio system, and computers, etc. 2.4 Rectify the faults diagnosed in 2.3 above using appropriate tools and instruments.	- Request students to perform diagnosis on some equipment	
	General Objective 3.0: Demonstrate the skill in alignment of electronic equipment		
WEEK	Specific Learning Outcome	Teachers Activities	Resources
13-15	3.1 State the various stages of the block diagram of the following: a. radio receivers b. Tv. Receivers c. Video cassette recorders d. Video camera e. Video monitors 3.2 Identify the various alignment methods for electronic equipment in 3.1 above. 3.3 Carry-out alignment on electronic equipment in 3.1	Demonstrate alignment on electronic equipment	
	ASSESSMENT: Course work 10%; Course tests 20%; Practical 20%; Examination 60%.		

USE OF BASIC MEASURING INSTRUMENTS

OSCILLOSCOPE

1. What is an oscilloscope?

An oscilloscope is a sophisticated diagnostic instrument that draws a graph of an electrical signal. This graph can tell you many things about a signal, such as: The time and voltage values of a signal. The frequency of an oscillating signal.

2. What Can Scopes Measure

In addition to those fundamental features, many scopes have measurement tools, which help to quickly quantify frequency, amplitude, and other waveform characteristics. In general, a scope can measure both time-based and voltage-based characteristics:

3. Timing characteristics:

I. Frequency and period -- Frequency is defined as the number of times per second a waveform repeat. And the period is the reciprocal of that (number of seconds each repeating waveform takes). The maximum frequency a scope can measure varies, but it's often in the 100's of MHz (1E6 Hz) range.

ii.Duty cycle -- The percentage of a period that a wave is either positive or negative (there are both positive and negative duty cycles). The duty cycle is a ratio that tells you how long a signal is "on" versus how long it's "off" each period.

iii. Rise and fall time -- Signals can't instantaneously go from 0V to 5V, they have to smoothly rise. The duration of a wave going from a low point to a high point is called the rise time, and fall time measures the opposite. These characteristics are important when considering how fast a circuit can respond to signals.

4. Voltage characteristics:

i.Amplitude -- Amplitude is a measure of the magnitude of a signal. There are a variety of amplitude measurements including peak-to-peak amplitude, which measures the absolute difference between a high and low voltage point of a signal. Peak amplitude, on the other hand, only measures how high or low a signal is past 0V.

ii. Maximum and minimum voltages -- The scope can tell you exactly how high and low the voltage of your signal gets.

iii. Mean and average voltages -- Oscilloscopes can calculate the average or mean of your signal, and it can also tell you the average of your signal's minimum and maximum voltage.

5. When to Use an O-Scope

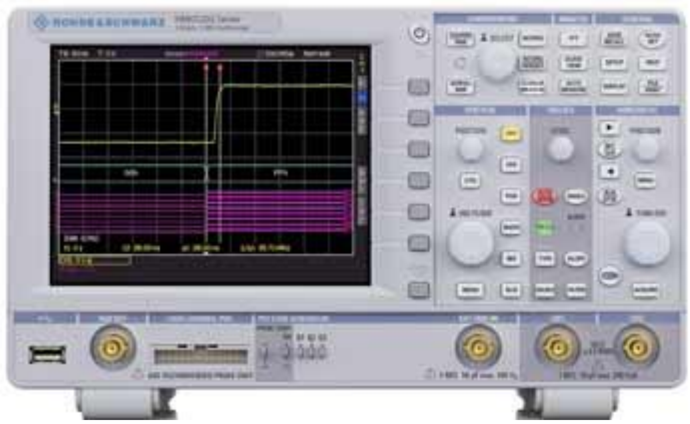
The o-scope is useful in a variety of troubleshooting and research situations, including:

i. Determining the frequency and amplitude of a signal, which can be critical in debugging a circuit's input, output, or internal systems. From this, you can tell if a component in your circuit has malfunctioned.

ii.Identifying how much noise is in your circuit.

iii.Identifying the shape of a wave -- sine, square, triangle, sawtooth, complex, etc.

iv.Quantifying phase differences between two different signals.



Basic oscilloscope controls

A summary of the main controls on an oscilloscope is given below:

I.Vertical gain: This control on the oscilloscope alters the gain of the amplifier that controls the size of the signal in the vertical axis. It is generally calibrated in terms of a certain number of volts per centimetre. Therefore by setting the vertical gain switch so that a lower number of volts per centimetre is selected, then the vertical gain is increased and the amplitude of the visible waveform on the screen is increased.

II.Vertical position: This control on the oscilloscope governs the position of trace when no signal is present. It is normally set to a convenient line on the graticule so that measurements above and below the "zero" position can be measured easily. It also has an equivalent horizontal position control that sets the horizontal position. Again this one should be set to a convenient position for making any timing measurements.

III.Timebase: The timebase control sets the speed at which the screen is scanned. It is calibrated in terms of a certain a certain time for each centimetre calibration on the screen. From this the period of a waveform can be calculated. This if a full cycle of a waveform too 10 microseconds to complete, this means that its period is 10 microseconds, and the frequency is the reciprocal of the time period, i.e. 1 / 10 microseconds = 100 kHz.

IV.Trigger: The trigger control on the oscilloscope sets the point at which the scan on the waveform starts. On analogue oscilloscopes, only when a certain voltage level had been reached by the waveform would the scan start. This would enable the scan on the waveform to start at the same time on each cycle, enabling a steady waveform to be displayed. By altering the trigger voltage, the scan can be made to start at a different point on the waveform. It is also possible to choose whether to trigger the oscilloscope on a positive, or a negative going part of the waveform. This may be provided by a separate switch marked with + and - signs.

steps in using an oscilloscope

Turn power on: This may appear obvious but is the first step. Usually the switch will be labelled "Power" or "Line". Once the power is on, it is normal for a power indicator or line indicator light to come on. This shows that power has been applied.

Wait for oscilloscope display to appear: Although many oscilloscopes these days have semiconductor-based displays, many of the older ones still use cathode ray tubes (crts), and these take a short while to warm up before the display appears. Even modern semiconductor ones often need time for their electronics to "boot-up". It is therefore often necessary to wait a minute or so before the oscilloscope can be used.

Find the trace: Once the oscilloscope is ready it is necessary to find the trace. Often it will be visible, but before any other waveforms can be seen, this is the first stage. Typically, the trigger can be set to the center and the hold-off turned fully counter-clockwise. Also set the horizontal and vertical position controls to the center, if they are not already there. Usually, the trace will become visible. If not the "beam finder" button can be pressed and this will locate the trace.

Set the gain control: The next stage is to set the horizontal gain control. This should be set so that the expected trace will nearly fill the vertical screen. If the waveform is expected to be 8 volts peak to peak, and the calibrated section of the screen is 10 centimeters high, then set the gain so that it is 1 volt / centimeter. This way the waveform will occupy 8 centimeters, almost filling the screen.

Set the time base speed: It is also necessary to set the time base speed on the oscilloscope. The actual setting will depend on what needs to be seen. Typically, if a waveform has a period of 10 ms and the screen has a width of 12 centimeters, then a time base speed of 1 ms per centimeter or division would be chosen.

Apply the signal: With the controls set approximately correctly the signal can be applied and an image should be seen.

Adjust the trigger: At this stage it is necessary to adjust the trigger level and whether it triggers on the positive or negative going edge. The trigger level control will be able to control where on the waveform the time base is triggered and hence the trace starts on the waveform. The choice of whether it triggers on the positive or negative going edge may also be important. These should be adjusted to give the required image.

Adjust the controls for the best image: With a stable waveform in place, the vertical gain and time base controls can be re-adjusted to give the required image.

SIGNAL(FUNCTION) GENERATOR

A signal generator is one of a class of electronic devices that generates electronic signals with set properties of amplitude, frequency, and wave shape. These generated signals are used as a stimulus for electronic measurements, typically used in designing, testing, troubleshooting, and repairing electronic or electroacoustic devices.

A signal generator may be as simple as an oscillator with calibrated frequency and amplitude.

Figure 1 function generator

Operation

Use the amplitude control to vary the voltage difference between the peaks and the troughs of the output signal. The frequency control varies the rate at which the signal oscillates. You need to first select a broad range of frequencies using the buttons on the left, and then select the frequency more precisely using the dial.

Output

Examples of sine, square, triangular and sawtooth waves are shown in figure 2

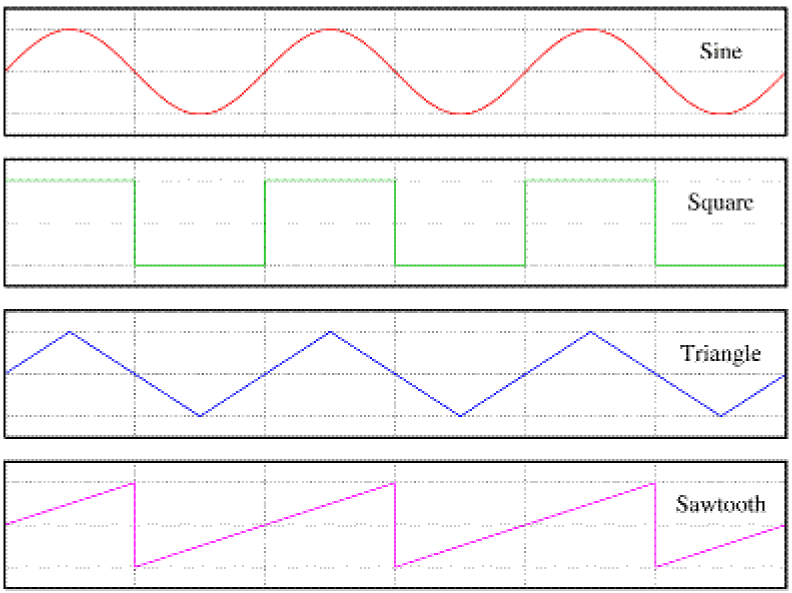


Figure 2 examples of square wave, triangular wave and sawtooth waves

MULTIMETER

A multimeter is a measuring instrument that can measure multiple electrical properties. A typical multimeter can measure voltage, resistance, and current, in which case it is also known as a volt-ohm-milliammeter (VOM), as the unit is equipped with voltmeter, ammeter, and ohmmeter functionality, However, some multimeters have extended measuring capability able to measure capacitance, inductance, hfe and continuity.

The most basic things we measure are voltage and current. A multimeter is also great for some basic sanity checks and troubleshooting. Is your circuit not working? Does the switch work? Put a meter on it! The multimeter is your first defense when troubleshooting a system. We can use the average multimeter to measure voltage, current, resistance, capacitance, Hfe and continuity.

Display

Selection Knob

Ports

The display usually has four digits and the ability to display a negative sign. A few multimeters have illuminated displays for better viewing in low light situations.

The selection knob allows the user to set the multimeter to read different things such as milliamps (mA) of current, voltage (V) and resistance (Ω).

Two probes are plugged into two of the ports on the front of the unit. COM stands for common and is almost always connected to Ground or '-' of a circuit. The COM probe is conventionally black but there is no difference between the red probe and black probe other than color. 10A is the special port used when measuring large currents (greater than 200mA). mAVΩ is the port that the red probe is conventionally plugged in to. This port allows the measurement of current (up to 200mA), voltage (V), and resistance (Ω). The probes have a banana type connector on the end that plugs into the multimeter. Any probe with a banana plug will work with this meter. This allows for different types of probes to be used

Probe Types

There are many different types of probes available for multimeters. Here are a few of our favorites:

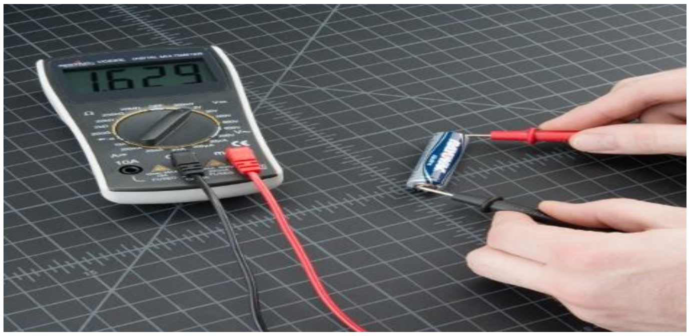
- Banana to Alligator Clips : These are great cables for connecting to large wires or pins on a breadboard. Good for performing longer term tests where you don't have to have to hold the probes in place while you manipulate a circuit.
- Banana to IC Hook : IC hooks work well on smaller ICs and legs of ICs.
- Banana to Tweezers : Tweezers are handy if you are needing to test SMD components.
- Banana to Test Probes : commonest



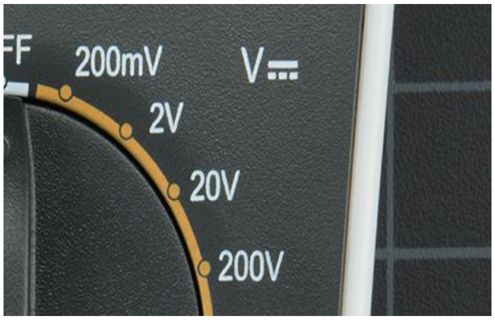

Banana to IC hook Banana to alligator Banana to test probe Banana to tweezer

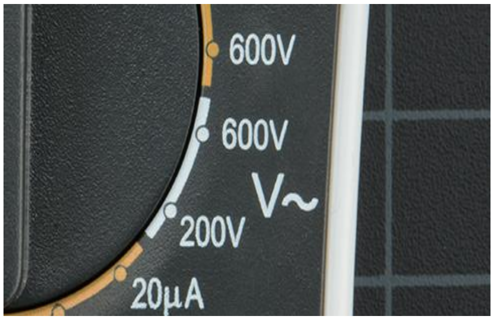
Measuring Voltage

To start, let's measure voltage on an AA battery: Plug the black probe into COM and the red probe into mAVΩ. Set the multimeter to "2V" in the DC (direct current) range. Almost all portable electronics use direct current), not alternating current. Connect the black probe to the battery's ground or '-' and the red probe to power or '+'. Squeeze the probes with a little pressure against the positive and negative terminals of the AA battery. If you've got a fresh battery, you should see around 1.5V on the display (this battery is brand new, so its voltage is slightly higher than 1.5V).



If you're measuring DC voltage (such as a battery or a sensor hooked up to an Arduino) you want to set the knob where the V has a straight line. AC voltage (like what comes out of the wall) can be dangerous, so we rarely need to use the AC voltage setting (the V with a wavy line next to it). If you're messing with AC, we recommend you get a non-contact tester rather than use a digital multimeter





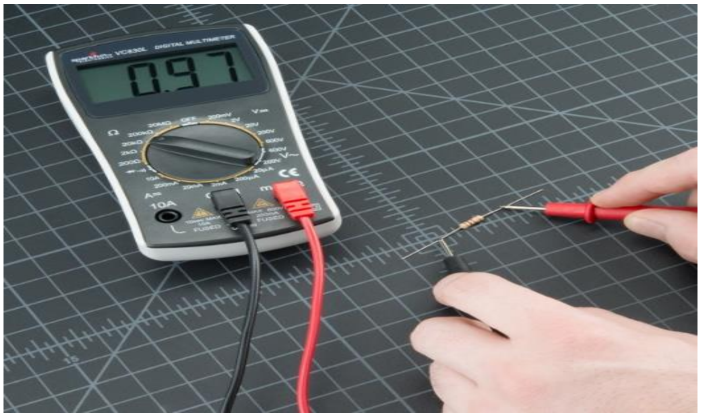
Use the V with a straight line for DC Voltage Use the V with a wavy line for AC Voltage

What happens if you switch the red and black probes? The reading on the multimeter is simply negative. Nothing bad happens! The multimeter measures voltage in relation to the common probe. How much voltage is there on the '+' of the battery compared to common or the negative pin? 1.5V. If we switch the probes, we define '+' as the common or zero point. How much voltage is there on the '-' of the battery compared to our new zero? -1.5V!

Measuring Resistance

Normal resistors have color codes on them. If you don't know what they mean, that's ok! There are plenty of online calculators that are easy to use. However, if you ever find yourself without internet access, a multimeter is very handy at measuring resistance.

Pick out a random resistor and set the multimeter to the 20kΩ setting. Then hold the probes against the resistor legs with the same amount of pressure you when pressing a key on a keyboard



the meter will read one of three things, 0.00, 1, or the actual resistor value.

- In this case, the meter reads 0.97, meaning this resistor has a value of 970Ω, or about 1kΩ (remember you are in the 20kΩ or 20,000 Ohm mode so you need to move the decimal three places to the right or 970 Ohms).
- If the multimeter reads 1 or displays OL, it's overloaded. You will need to try a higher mode such as 200kΩ mode or 2MΩ (megaohm) mode. There is no harm if this happen, it simply means the range knob needs to be adjusted.
- If the multimeter reads 0.00 or nearly zero, then you need to lower the mode to 2kΩ or 200Ω.

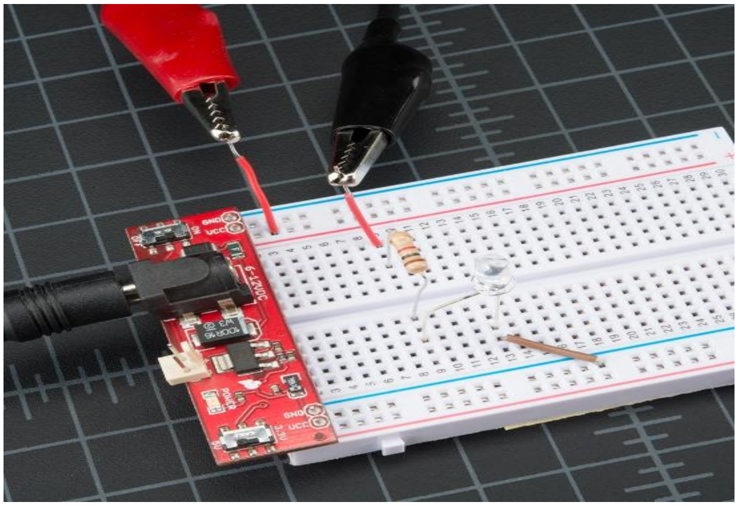
Remember that many resistors have a 5% tolerance. This means that the color codes may indicate 10,000 Ohms (10kΩ), but because of discrepancies in the manufacturing process a 10kΩ resistor could be as low as 9.5kΩ or as high as 10.5kΩ. Don't worry, it'll work just fine as a pull-up or general resistor.

Measuring Current

Reading current is one of the trickiest and most insightful readings in the world of embedded electronics. It's tricky because you have to measure current in series. Where voltage is measure by poking at VCC and GND (in parallel), to measure current you have to physically interrupt the flow of current and put the meter in-line. To demonstrate this, we'll use the same circuit we used in the measuring voltage section.

The first thing we'll need is an extra piece of wire. As mentioned, we'll need to physically interrupt the circuit to measure the current. Said another way, pull out the VCC wire going to the resistor, add a wire where that wire was connected, and then probe from the power pin on the power supply to the resistor. This effectively "breaks" power to the circuit. We then insert the multimeter in-line so that it can measure the current as it "flows" through to the multimeter into the bread board.

For these pictures, we cheated and used alligator clips. When measuring current, it's often good to watch what your system does over time, for a few seconds or minutes. While you might want to stand there and hold the probes to the system, sometimes it's easier to free up your hands. These alligator clip probes can come in handy. Note that almost all multimeters have the same sized jacks (they're called "banana plugs") so if you're in a pinch, you can use your friend's probes.



With the multimeter connected, we can now set the dial to the proper setting and measure some current. Measuring current works the same as voltage and resistance -- you have to get the correct range. Set the multimeter to 200mA, and work from there. The current consumption for many breadboard projects is usually under 200mA. Make sure the red probe is plugged into the 200mA fused port. On our favorite multimeter, the 200mA hole is the same port/hole as voltage and resistance reading (the port is labeled mAVΩ). This means you can keep the red probe in the same port to measure current, voltage, or resistance. However, if you suspect that your circuit will be using close to or more than 200mA, switch your probe to the 10A side, just to be safe. Overloading the current can result in a blown fuse rather than just an overload display. More on that in a bit.

Continuity

Continuity testing is the act of testing the resistance between two points. If there is very low resistance (less than a few Ωs), the two points are connected electrically, and a tone is emitted. If there is more than a few Ωs of resistance, than the circuit is open, and no tone is emitted. This test helps ensure that connections are made correctly between two points. This test also helps us detect if two points are connected that should not be.

Continuity is quite possibly the single most important function for embedded hardware gurus. This feature allows us to test for conductivity of materials and to trace where electrical connections have been made or not made.

Set the multimeter to 'Continuity' mode. It may vary among DMMs, but look for a diode symbol with propagation waves around it (like sound coming from a speaker).

Now touch the probes together. The multimeter should emit a tone (Note: Not all multimeters have a continuity setting, but most should). This shows that a very small amount of current is allowed to flow without resistance (or at least a very small resistance) between probes.

On a breadboard that is *not* powered, use the probes to poke at two separate ground pins. You should hear a tone indicating that they are connected. Poke the probes from the VCC pin on a microcontroller to VCC on your power supply. It should emit a tone indicating that power is free to flow from the VCC pin to the micro. If it does not emit a tone, then you can begin to follow the route that copper trace takes and tell if there are breaks in the line, wire, breadboard, or PCB.

Continuity is a great way to test if two SMD pins are touching. If your eyes can't see it, the multimeter is usually a great second testing resource.

When a system is not working, continuity is one more thing to help troubleshoot the system. Here are the steps to take:

1. If the system is on, carefully check VCC and GND with the voltage setting to make sure the voltage is the correct level. If the 5V system is running at 4.2V check your regulator carefully, it could be very hot indicating the system is pulling too much current.
2. Power the system down and check continuity between VCC and GND. If there is continuity (if you hear a beep), then you've got a short somewhere.
3. Power the system down. With continuity, check that VCC and GND are correctly wired to the pins on the microcontroller and other devices. The system may be powering up, but the individual ICs may be wired wrong.
4. Assuming you can get the microcontroller running, set the multimeter aside, and move on to serial debugging or use a logic analyzer to inspect the digital signals.

Continuity and large capacitors: During normal troubleshooting. you will be probing for continuity between ground and the VCC rail. This is a good sanity check before powering up a prototype to make sure there is not a short on the power system. But don't be surprised if you hear a short 'beep!' when probing. This is because there are often significant amounts of capacitance on the power system. The multimeter is looking for very low resistance to see if two points are connected. Capacitors will act like a short for a split second until they fill up with energy, and then act like an open connection. Therefore, you will hear a short beep and then nothing. That's ok, it's just the caps charging up.

Measuring Capacitance

Use your digital multimeter (DMM) to ensure all power to the circuit is OFF. If the capacitor is used in an ac circuit, set the multimeter to measure ac voltage. If is used in a dc circuit, set the DMM to measure dc voltage.

Visually inspect the capacitor. If leaks, cracks, bulges or other signs of deterioration are evident, replace the capacitor.

Turn the dial to the Capacitance Measurement mode. The symbol often shares a spot on the dial with another function. In addition to the dial adjustment, a function button usually needs to be pressed to activate a measurement. Consult your multimeter's user manual for instructions..

5. For a correct measurement, the capacitor will need to be removed from the circuit. Discharge the capacitor as described in the warning above.

PREACUTIONS TO BE TAKEN WHILE TESTING

General safety guidelines

Use the following safety guidelines to protect your equipment from potential damage and to ensure your personal safety. Unless otherwise noted, each procedure included in this document assumes that you have read the safety information that shipped with your computer.

1. Place the equipment on a hard, level surface. Leave a 10.2 cm (4 in) minimum of clearance on all vented sides of the computer to permit the airflow required for proper ventilation. Restricting airflow can damage the computer or cause a fire.
2. Do not stack equipment or place equipment so close together that it is subject to re-circulated or preheated air.
3. Ensure that nothing rests on your equipment's cables and that the cables are not located where they can be stepped on or tripped over.
4. Ensure that all cables are connected to the appropriate connectors. Some connectors have a similar appearance and may be easily confused (for example, do not plug a telephone cable into the network connector).
5. Do not place your device in a closed-in wall unit, or on a soft, fabric surface such as a bed, sofa, carpet, or a rug.
6. Keep your device away from radiators and heat sources.
7. Keep your equipment away from extremely hot or cold temperatures to ensure that it is used within the specified operating range.
8. Do not push any objects into the air vents or openings of your equipment. Doing so can cause fire or electric shock by shorting out interior components.
9. Do not use your equipment in a wet environment, for example, near a bath tub, sink, or swimming pool or in a wet basement.
10. Do not use AC powered equipment during an electrical storm. Battery powered devices may be used if all cables have been disconnected.
11. Do not spill food or liquids on your equipment.
12. Before you clean your equipment, disconnect it from the electrical outlet. Clean your device with a soft cloth dampened with water. Do not use liquids or aerosol cleaners, which may contain flammable substances.
13. Clean LCD screen or monitor display with a soft, clean cloth and water. Apply the water to the cloth, then stroke the cloth across the display in one direction, moving from the top of the display to the bottom. Remove moisture from the display quickly and keep the display dry. Long-term exposure to moisture can damage the display. Do not use a commercial window cleaner to clean your display.

General power safety guidelines

- Check the voltage rating before you connect the equipment to an electrical outlet to ensure that the required voltage and frequency match the available power source.
- Your device is equipped with either an internal power source or an external adapter. For internal power sources, your device is equipped with one of the following:
  - An auto-sensing voltage circuit – Devices with an auto-sensing voltage circuit do not have a voltage selection switch on the back panel and automatically detect the correct operating voltage.
  - Or, a manual voltage selection switch – Devices with a voltage selection switch on the back panel must be manually set to operate at the correct operating voltage. Set the switch to the position that most closely matches the voltage used in your location
- Ensure that the equipment is electrically rated to operate with the AC power available in your location.
- Do not plug the equipment power cables into an electrical outlet if the power cable is damaged.
- To prevent electric shock, plug the equipment power cables into properly grounded electrical outlets. If the equipment is provided with a 3-prong power cable, do not use adapter plugs that bypass the grounding feature, or remove the grounding feature from the plug or adapter.
- If you use an extension power cable, ensure that the total ampere rating of the products plugged in to the extension power cable does not exceed the ampere rating of the extension cable.
- If you must use an extension cable or power strip, ensure the extension cable or power strip is connected to a wall power outlet and not to another extension cable or power strip. The extension cable or power strip must be designed for grounded plugs and plugged into a grounded wall outlet.
- If you are using a multiple-outlet power strip, use caution when plugging the power cable into the power strip. Some power strips may allow you to insert a plug incorrectly. Incorrect insertion of the power plug could result in permanent damage to your equipment, as well as risk of electric shock and/or fire. Ensure that the ground prong of the power plug is inserted into the mating ground contact of the power strip.
- Be sure to grasp the plug, not the cable, when disconnecting equipment from an electric socket.
- Use only AC adapter approved devices. Use of another AC adapter may cause a fire or explosion.
- Place the AC adapter in a ventilated area, such as a desk top or on the floor, when you use it to run device or to charge the battery. Do not cover the AC adapter with papers or other items that will reduce cooling; also, do not use the AC adapter inside a carrying case.
- The AC adapter may become hot during normal operation of your devicer. Use care when handling the adapter during or immediately after operation.
- It is recommended that you lay the adapter on the floor or desk so that the green light is visible. This will alert you if the adapter should accidentally go off due to external effects. If for any reason the green light goes off, disconnect the AC cord from the wall for a period of ten seconds, and then reconnect the power cord.
- Never use an AC adapter that shows signs of damage or excessive wear.

ALIGNMENT

Alignment is the proper positioning or state of adjustment of parts (as of a mechanical or electronic device) in relation to each other for the purpose of obtaining best operation of the device.

Aligning the radio receiver

The alignment of an amplitude modulated radio receiver consists of two types of adjustments of the different tuned circuits present in the receiver. The first part is to align the I.F. transformers to the correct intermediate frequency of the receiver. This is called the I.F. alignment. The second part is to align the input tuned circuits to match with the scale/dial markings such that the wanted station (frequency) is producing the correct value of I.F. when tuned on the dial to the frequency marked on the dial. This is called the R.F. alignment. The R.F. alignment is also called TRACKING.

The block diagram for IF alignment is given as

SET UP FOR I.F. ALIGNMENT OF RADIO RECEIVER

To track the radio receiver a standard signal generator is to be tuned to the desired frequency as listed here under. The signal will not be directly connected to the input of the receiver, but the probe will be kept close to the antenna coils of the receiver. The frequencies to which the standard signal generator and the receiver are to be tuned and the adjustment to be made are given step by step here under in Table.

S.G. Frequency	Position of pointer	Adjustment to be made for getting maximum output
SWITCH THE RECEIVER TO THE MEDIUM WAVE BAND (550 kHz to 1600 kHz)		
550 kHz	550 kHz	M.W. Oscillator coil
1600 kHz	1600 kHz	Trimmer of M.W. Osc. Coil
Repeat this adjustment twice then:		
840 kHz	840 kHz	Slide the M.W. antenna coil on the ferrite rod.
Repeat the above two operations twice to get tuning as close to the dial marking as is possible.		
SWITCH THE RECEIVER TO THE SHORT WAVE BAND (4.5 MHz to 16 MHz)		
04.5 MHz	04.5 MHz	S.W. Oscillator Coil
16.5 MHz	16.5 MHz	Trimmer S.W. Osc. Coil
Repeat this adjustment twice then:		
05.0 MHz	05.0 MHz	S.W. Antenna Coil
Repeat the above two operations twice to get tuning as close to the dial marking as is possible.		

Aligning the CD player

Play adjustments

You will see a circuit board, hopefully in your unit it is readily accessible with component markings. For each servo, there will be 1 or 2 pots to adjust. Unfortunately for our purposes, some CD players have no adjustments! In this case about all you can do is confirm that the lens is clean and clean and lubricate the mechanism.

The adjustments will be labeled something like:

1. Focus - F.G. (focus gain), F.O. (focus offset)

2,3. Tracking - T.G. (tracking gain), T.O. (tracking offset), maybe others.

4. Spindle PLL, PLL adj., Speed, or something like that.

DO NOT TOUCH THE LASER POWER ADJUSTMENT - you can possibly ruin the laser if you turn it up too high. Sometimes, just turning it with power applied can destroy the laser diode due to a noisy potentiometer. This adjustment can only be made properly with the service manual. It may require an optical power meter to set laser output. Very often the adjustment is on the optical pickup itself so it should be easy to avoid. Sometimes it is on the main PCB. The laser optical power output is feedback controlled and unlikely to change unless the laser is defective - in which case adjustments will have little effect anyway. If you run out of options, see the section: "Laser power adjustment" - last.

DO NOT JUST GO AND TWEAK WILDLY. You will never be able to get back to a point where the disc will even be recognized (without test equipment and probably a service manual).

First, somehow mark the EXACT positions of each control. Some of these may require quite precise setting - a 1/16 of a turn could be critical, especially for the offset adjustments. Sometimes, there will be marked test points, but even then, the exact procedure is probably model dependent.

Adjustment procedure for noise or skipping

The assumption here is that you can get the disc to play but there is audio noise skipping, or other similar problem.

Play a disc at the track that sounds the worst - put it into repeat mode so it will continue for a while. Get it to play by whatever means works.

Repetitive noise at disc rotation frequency

Try to locate the adjustments for focus. Try the focus offset first, just a hair in each direction. If you go too far, you will lose focus lock totally, the servo will go into focus search mode and/or the unit will shut down. Return the control to the exact original position if there is no improvement. You can also try gain, but in my experience, the gain controls are not critical to normal play but determine how the unit will handle dirty and/or defective discs. However, if they are way off, there could be general problems. Too low a gain setting (this applies to focus as well as tracking) will make the unit very prone to skipping as a result of minor bumps. Too high a setting will make the unit skip as a result of minor disc defects.

Short distance skipping or sticking

Try to locate the adjustments for tracking. Try the fine tracking offset first, just a hair in each direction. If you go too far, you will lose servo lock totally, the pickup will slew to one end of the disc, and/or the unit will shut down. Return the control to the exact original position if there is no improvement. Then try the other tracking offset if there is one and also the gain (though this is probably not the problem).

Always return each control to its original position after the test so you don't confuse things more.

Aligning Satellite Dish

To get a satellite dish aligned correctly, you must maximize the signal on three plains. This is the left and right or east and west (azimuth) adjustment. The elevation angle which is up and down and the skew adjustment, this is angle that the LNB itself is fixed into the LNB holder.

Azimuth (East & West) adjustment

The azimuth is the most important alignment of them all as once you have got this fixed, it is just a case of setting the elevation before you start to find the satellite. If you have the azimuth adjustment off, you could waste hours finding the wrong satellites (there are loads) so you will need to do some investigation on the satellite angles that you wish to use.

Elevation Adjustment

Once you have the azimuth set correctly so will need to correctly set the elevation angle before you find the satellite, this is again given in an angle, but the absolute correct angle will depend on where you are on the earth. As the telecommunication satellites orbit around the equator, the Astra 2 satellites occupy a geo-stationary orbit position above the Democratic Republic of Congo for example the closer the equator you're the higher in the Sky the satellites will be in the horizon. Until you get to the equator where the satellites will literally be straight up above you, the other extreme of this will be in places like Iceland where the satellites will appear very low on the horizon which can make line of sight much trickier due to nearby obstructions like trees and nearby buildings.

Skew Adjustment/ LNB setting

The final alignment that needs to be set is the skew adjustment on the LNB. This is the angle that the LNB itself sits within the LNB holder. Most people that these should be set perfectly straight, I have even had a customer remark to me that the LNB was not installed straight but more often than not to install your LNB with a zero-degree angle on the skew adjustment will be wrong and really will affect the satellite signal reliability. Unless the satellites that you wish to align your dish to are perfectly due south (or north down under!) then you are going to need to make an adjustment for the LNB skew.

Tools To Help You Find the Satellites

1. Kingofsat – Information On Satellite TV Services

2. What Satellite Magazine (Wotsat) – Information On Dish Sizes

3. Dishpointer – Information On Satellite Angles

Dishpointer allows you to do is enter your geographic location, this can be anywhere on the planet and the satellites you wish to receive from, hence finding this out from Kingofsat first and this will give you the co-ordinates for your elevation angle and skew adjustment. It will also give you a very helpful Birdseye view map with a line in the direction of the satellite.

4. Compass/ Inclometer

5. Satellite Dish Alignment Tool – TV Spectrum Analyser

6. satellite Receiver Bleeper

Tips on How to Align the Dish

Use Neighbor's Satellite Dish As A Reference

One thing could save you a fair amount of time, certainly picking the correct position would be to use a neighbour's satellite dish as a reference point to where you want to point your own dish.

Avoid Trees

Although with the natural elevation setting the signals could be beamed comfortably over the top of nearby trees I would say try and avoid them altogether in the satellite line of sight. This may mean that you must install your satellite dish in a different position than you had intended, perhaps with a longer cable run but it could save you lots of bother having to move it at a later date. Trees can cause problems for several reasons:

1- They move – Branches could blow in front of the satellite signal

2- They grow – Even if you have a signal today, you might not in a summer of two when the tree has grown

3- They leaf in summer – If you are installing a satellite dish pointing through a bald tree, the signal may pass through this fine, but come the summer when the leaves are out, this will most likely be a different story